Pragati Engineering College
(Autonomous)

PROJECT :

Smart Personal Food Safety and Healthcare Assistant

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Smart Personal Food Safety and Healthcare Assistant

Abstract :

The Smart Personal Food Safety and Healthcare Assistant is a C++-based decision-support project designed to help users monitor food safety, understand nutrition, evaluate grocery quality, and make safer diet decisions for patients with specific health conditions. The system combines food expiry tracking, leftover safety checking, nutrient viewing, patient diet restriction matching, hospital contact support, and grocery quality assessment in a single menu-driven application.By bringing these functions together, the project aims to reduce food waste, lower the risk of foodborne illness, and improve day-to-day health awareness in homes and caregiving settings.

Problem Statement :

Many people store food without properly tracking expiry dates, leftover duration, nutrient value, or visible spoilage signs, which increases the risk of consuming unsafe food.Patients with conditions such as diabetes, hypertension, kidney disease, and food allergies also need food choices that match their medical restrictions, but such checks are not always available in a simple daily-use system. Existing nutrition support in clinical care is individualized through medical nutrition therapy, which shows the importance of condition-based dietary guidance, but there is still room for a basic personal decision-support tool that offers fast preliminary screening for users and caretakers.

Differences :

This project differs from a normal expiry tracker because it does more than store dates and display alerts. It combines food spoilage checking, nutrient awareness, patient diet suitability, grocery quality scoring, and emergency hospital contact support in one system,

making it broader than a standard pantry manager or hospital contact directory alone.It also differs from advanced commercial food scanner apps because it is designed as a C++ academic project using rule-based logic, file handling, and object-oriented programming rather than requiring a full mobile cloud ecosystem.

Problem Solutions :

The project addresses unsafe food consumption by checking expiry dates, leftover duration, and spoilage indicators such as odor, leakage, texture change, or visible contamination. It addresses patient dietary safety by comparing food nutrition and ingredient risks with the selected patient condition or allergy profile, helping the user decide whether a food is suitable, risky, or should be avoided. It also improves emergency readiness by storing hospital contact information and improves buying decisions by giving grocery items a quality score based on freshness and package condition.

USP’s

The project’s major unique selling point is that it combines food safety + nutrition support + patient diet guidance + grocery quality checking + hospital contact support in a single C++ application. A second USP is that it can be implemented completely in C++ with classes, vectors, decision rules, and file handling, making it practical for academic demonstration without depending on a web stack.A third USP is that it is socially useful because it can support households, elderly people, patients, and caretakers in making faster and safer food decisions.

List of Features :

- Food item entry with purchase date, expiry date, storage type, and quantity.
- Food spoil date check using current date and expiry date comparison.
- Leftover safety check based on storage time; USDA says leftovers should generally be eaten or frozen within four days.Safe-to-eat decision support using spoilage indicators such as bad smell, leakage, fungus, or color change.
- Nutrition information display including protein, vitamins, sugar, sodium, and calories.
- Patient health profile storage including disease type and allergy information.
- Patient food suitability checker for diabetes, hypertension, kidney disease, and allergy-sensitive cases.
- Grocery quality score based on freshness and package condition.
- Hospital and emergency contact management.
- File save/load support using text or CSV files in C++

Mock Diagram :

```
+-----------------------------------------------+
| Smart Personal Food Safety and Healthcare Assistant |
+-----------------------------------------------+
| 1. Add Food Item                              |
| 2. View Food Items                            |
| 3. Check Spoilage / Expiry                    |
| 4. Check Safe to Eat                          |
| 5. View Nutrition Details                     |
| 6. Add Patient                                |
| 7. Check Patient Can Eat or Not               |
| 8. Grocery Quality Check                      |
| 9. Hospital Contacts                          |
| 10. Save / Load Data                          |
| 11. Exit                                      |
+-----------------------------------------------+
```

Architecture Diagram :

```
┌──────────────────────────┐
│      User Input          │
└──────────────────────────┘
            │
            ▼
┌──────────────────────────┐
│    Main Application      │
└──────────────────────────┘
        │   │   │
        ▼   ▼   ▼
┌────────────────┐ ┌────────────────┐ ┌──────────────────────┐
│  Food Module   │ │Patient Module  │ │  Hospital Module     │
│FoodItem class  │ │Patient class   │ │HospitalContact       │
└────────────────┘ └────────────────┘ └──────────────────────┘
        │                │                      │
        ▼                ▼                      ▼
┌────────────────┐ ┌────────────────┐ ┌──────────────────────┐
│ Safety Engine  │ │  Diet Engine   │ │ Contact Manager      │
│Expiry check    │ │Can eat/avoid   │ │Search/display        │
│Spoilage logic  │ │Allergy/disease │ │Emergency info        │
└────────────────┘ └────────────────┘ └──────────────────────┘
        │                │
        └────────────────┘
                │
                ▼
        ┌──────────────────────┐
        │    File Manager      │
        │ Text/CSV files       │
        └──────────────────────┘
```

Technology used :

Component	Technology
Programming Language	C++
Programming Style	Object-Oriented Programming with classes and modular file
Data Structures	`vector`, `string`, `struct/class`, arrays where required
File Storage	Text files / CSV using `fstream`
Logic Type	Rule-based decision system for expiry, safety, and patient diet suitability
Platform	VS Code or any C++ compiler environment

Conclusion :

The Smart Personal Food Safety and Healthcare Assistant is a practical and innovative C++ project that expands a normal expiry tracker into a broader food and health decision-support system. By integrating spoilage checking, nutrient awareness, patient suitability analysis, hospital contact support, and grocery quality evaluation, the system becomes more useful, more realistic, and more valuable for academic presentation. The project should be presented as a rule-based guidance tool rather than a replacement for doctors or dietitians, because medical nutrition decisions are individualized in professional care